

Group counting expressions as a window into the internal structure of numerals

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<https://bit.ly/numerals-dm-nano>



This talk

Abstract use of numerals

(1) **Two** is my favorite number.

Counting use of numerals

(2) Eve ate **two** apples.

Previous work: numeral counting expressions like (2) properly contain abstract number concept expressions like (1). (Wągiel and Caha 2020; Wągiel and Caha 2021)

This talk: Also number concept expressions like (1) are **internally complex**.

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This talk: Also number concept expressions like (1) are **internally complex**.

Data

- ▶ Understudied numeral constructions: **group counting** expressions

(3) **three sets** of cards

- ▶ Taking into account group counting constructions yields paradigms that provide insights into internal make-up of numerals.

⇒ Our findings inform debates on the relation between abstract number concepts, unit and group counting in natural language. (E.g. Bale, Gagnon, et al. 2011; Bale and Coon 2014; Bale, Coon, and Arcos 2019; Ionin and Matushansky 2006; Grimm and Dočekal 2021; Sağ 2018; Rothstein 2017)

Roadmap

Unit vs. group counting

Pattern 1: unit $\rightarrow$ bimorphemic

Pattern 2: unit $\rightarrow$ bimorphemic

group $\rightarrow$ bimorphemic

group $\rightarrow$ tri-morphemic

Abstract use of numerals

Abstract number concepts

Pattern 1: abstract bare *Num*

Pattern 2: abstract *Num+A*

Summary of patterns

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Mandarin and Uzbek

Yucatec Maya and Bwamu

Conclusion

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Unit vs. group counting

Unit counting use of numerals

- ▶ By far most studied type of numeral construction
- ▶ Numerals counting **NP-atoms**
- ▶ (4): set of pluralities each consisting of five apple atoms

(Landman 2003; Krifka 1989)

(4) **five** apples

Unit vs. group counting

Unit counting use of numerals

- ▶ By far most studied type of numeral construction
- ▶ Numerals counting **NP-atoms**
- ▶ (4): set of pluralities each consisting of five apple atoms
(Landman 2003; Krifka 1989)

(4) **five** apples

Group counting use of numerals

- ▶ Counting of **NP-non-atoms**
- ▶ (5): set of pluralities each consisting of five woman non-atoms
(Landman 1989; Dočekal 2012; Grimm and Dočekal 2021; Khrizman 2020)

(5) **five groups of** women

- ▶ Other languages: dedicated group counting numerals or classifiers
 - **Num_{gr} + NP** ▪ **Num + CLF_{gr} + NP**

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Unit vs. group counting: Pattern 1

Mandarin

(6) a. sān **tóu** niú
three CLF₁ cow
'three cows'

b. sān **qún** niú
three CLF₂ cow
'three herds of cows'

Uzbek

(7) a. ikki **dona** kitob
two CLF₁ book
'two books'

b. ikki **guruh** bola
two CLF₂ child
'two groups of children'

Unit vs. group counting: Pattern 1

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Unit vs. group counting: Pattern 1

Pattern 1

Unit counting: **Num** + **A** 2-morphemic

Group counting: **Num** + **B** 2-morphemic

- ▶ Seems to be prevalent pattern in most classifier languages
- ▶ E.g. Chinese *three*, Uzbek *two*

NB: Not all numerals necessarily behave the same within a language.

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Unit vs. group counting: Pattern 2

Bwamu (Gur)

- (8) a. livru-ʒε **bo** tĩ
book-PL CLF₁ three
'three books'
- b. hã-ʒε **gnũ-nε** **bo** tĩ
woman-PL CLF₂-PL CLF₁ three
'three groups of women'

Yucatec Maya

- (9) a. jum-p'éeł kib
one-CLF₁ candle
'one candle'
- b. jun-p'èel **cháach** iib
one-CLF₁ CLF₂ bean
'one bunch of beans'

Unit vs. group counting: Pattern 2

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Unit vs. group counting: Pattern 2

Finnish

- (10) a. **vii-si** saapasta
 five-SFX boot.PTV
 'five-SFX₁ boots'
- b. **vii-de-t** saappaat
 five-SFX₁-SFX₂ boots
 'five pairs of boots'

Czech

- (11) a. **tř-i** karty
 three-SFX cards
 'three cards'
- b. **tr-oj-e** karty
 three-SFX₁-SFX₂ cards
 'three sets of cards'

Unit vs. group counting: Pattern 2

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- (10) a. NUM A
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- b. tři-**o**j-e karty
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[Decomp. of *-oj-e* and *-de-t*: motivation in Appendix]

Unit vs. group counting: Pattern 2

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- (10) a. NUM A
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- b. NUM B C
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Unit vs. group counting: Pattern 2

Pattern 2

Unit counting:

Num + **A**

2-morphemic

Group counting:

Num + **A** + **C**

3-morphemic

(Bwamu, Y.Maya)

Num + **B** + **C**

3-morphemic

(Czech, Finnish)

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Abstract use of numerals

- ▶ Focus in semantic literature: numerals in their **counting function**
(e.g. Bale, Gagnon, et al. 2011, Martí 2020a; Martí 2020b, Scontras 2013, Saĝ 2018)
- ▶ Numerals with NP complements: **Num + NP**.
- ▶ Device for counting individuals in the NP denotation.
- ▶ Pluralities of NP individuals.

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Abstract use

- ▶ Non-counting contexts, e.g. maths, reference to number concepts
- ▶ English: only **bare numerals**

- (12) a. Four times two is eight.
 b. My favorite number is one.

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- (12) a. **Four** times **two** is eight.
b. My favorite number is **one**.

- ▶ Even semantically inert nouns cannot occur here
- ▶ Shows: these contexts select for items denoting **abstract number concepts**
(e.g. Rothstein 2017; see also Moltmann 2013; Moltmann 2017; Wągiel and Caha 2020; Wągiel and Caha 2021)

- (13) a. #**Four** things times **two** things is **eight** things.
b. #My favorite number is **one** thing.

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- (13) a. #**Four** things times **two** things is **eight** things.
b. #My favorite number is **one** thing.

- ⇒ Semantics of numerals is **not inherently** that of **counting** NP denotations
- ⇒ Certain instances of numerals denote **abstract number concepts**, not counting devices.

We will show

Pattern 1

Mandarin, Uzbek

Unit: **Num** + **A** 2-morph

Group: **Num** + **B** 2-morph

Abstract use

Num

Pattern 2

Bwamu, Y.Maya, Czech, Finnish

Unit: **Num** + **A** 2-morph

Group: **Num** + **A** + **C** 3-morph

Num + **B** + **C**

Abstract use

Num + **A**

We will show

Pattern 1

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Unit: **Num** + **A** 2-morph

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Bwamu, Y.Maya, Czech, Finnish

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Num + **B** + **C**

Abstract use

Num + **A**

⇒ What appears as **Num** in Pattern 1 realizes **two** features.

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Pattern 1: abstract bare *Num*

Mandarin

- (14) a. **jiǔ** chùyǐ **sān** shì **sān**.
nine divide.by three COP three
'Nine divided by three is three.'
- b. #**jiǔ** **gè** chùyǐ **sān** **gè** shì **sān** **gè**.
nine **CLF** divide.by three **CLF** COP three **CLF**

Uzbek (Levy-Forsythe 2025, p. 203)

- (15) a. **Ikki** karra **uch** barobar **olit**.
two times three equal six
'Two times three equals six.'
- b. #**Ikki-ta** karra **uch-ta** barobar **olit-ta**.
two-**CL_{unit}** times three-**CL_{unit}** equal six-**CL_{unit}**

- ⇒ **No classifiers** in abstract use of numerals
- ⇒ Classifiers in these languages are for **counting**

Full pattern 1

Pattern 1

Abstract:	Num	1-morphemic
Unit counting:	Num + A	2-morphemic
Group counting:	Num + B	2-morphemic

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Pattern 2: abstract *Num* + *A*

Bwamu (Bouwèrè Bognana, p.c.)

- (16) a. **bo** na la **bo** tĩ le **bo** hognu
CLF four PRE CLF three COP CLF seven
'Four plus three equals seven.'
- b. #na la tĩ le hognu
four PRE three COP seven

Yucatec Maya (Blaha Pfeiler and Skopeteas 2024)

- (17) a. **jum-p'éel** y-éetel u láak' **ka'a-p'éel-e'** **óox-p'éel**.
one-CLF_{unit} Ø-with A.3 other two-CLF_{unit}-TOP three-CLF_{unit}
'One and two is three.'
- b. #**jum** y-éetel u láak' **ka'a-e'** **óox**.
one-CLF_{unit} Ø-with A.3 other two-CLF_{unit} TOP three-CLF_{unit}

- ⇒ Numerals denoting abstract number concepts are **bi-morphemic**.
⇒ They need to contain a 'classifier'.

Pattern 2: abstract *Num* + *A*

Czech

- (18) a. Tř-i je prvočíslo.
three-SFX is prime.number
'Three is a prime number.'
- (19) a. #Tř-∅ je prvočíslo.
three-∅ is prime.number

Finnish (Karlsson 2017 p.300)

- (20) kak-si plus kak-si on neljä
two plus two is four
'two plus two is four'
- (21) a. #kak-∅ plus kak∅ on neljä-t
two-∅ plus two-∅ is four-PL

⇒ Numerals denoting abstract number concepts are **bi-morphemic**.

Full Pattern 1 and 2

Pattern 1

Abstract:	Num	1-morphemic
Unit counting:	Num + A	2-morphemic
Group counting:	Num + B	2-morphemic

Pattern 2

Abstract:	Num + A	2-morphemic	
Unit counting:	Num + A	2-morphemic	
Group counting:	Num + A + C	3-morphemic	(Bwamu, Y.Maya)
	Num + B + C	3-morphemic	(Czech, Finnish)

⇒ **Abstract number concept** denoting numerals are **internally complex** and composed of two features.

⇒ This complexity is **covert in Pattern 1** but **overt in Pattern 2**.

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Pattern 1 summary

Mandarin 'three'

san

san **tóu**_{unit}

san **qún**_{group}

Num

Num + **A**

Num + **B**

abstract

unit

group

Uzbek 'two'

ikki

ikki **dona**_{unit}

ikki **guruh**_{group}

Num

Num + **A**

Num + **B**

abstract

unit

group

Pattern 2 summary

Bwamu 'three'

bo tĩ

bo tĩ

gnũ-ne bo tĩ

Num + A

Num + A

Num + A + B

abstract

unit

group

Yucatec Maya 'one'

jum-p'éel

jum-p'éel

jum-p'éel cháach

Num + A

Num + A

Num + A + C

abstract

unit

group

Czech 'three'

tř-i

tř-i

tr-oj-e

Num + A

Num + A

Num + B + C

abstract

unit

group

Finnish 'two'

kak-si

kak-si

kah-de-t

Num + A

Num + A

Num + B + C

abstract

unit

group

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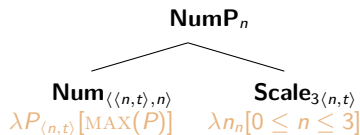
Appendix

Representation of abstract number concepts

Representation of abstract number concepts

- ▶ Number concept expressions realize **two features**: [Num] and [Scale]
- ▶ (22): structure of abstract number concept expressions
- ▶ As in *My favorite number is **three***.

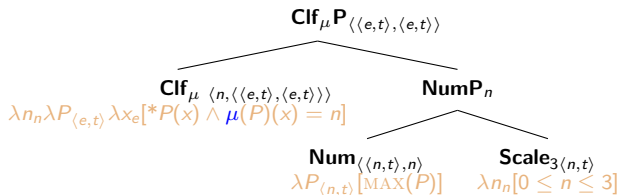
(22)



Representation of numerals as counting devices

- ▶ Classifiers turn abstract number concepts into **counting devices**
- ▶ Two distinct Clf heads:
 - ▶ $\llbracket \mathbf{Clf}_{\text{unit}} \rrbracket = \lambda n_n \lambda P_{\langle e, t \rangle} \lambda x_e [*P(x) \wedge \#(P)(x) = n]$
 - ▶ $\llbracket \mathbf{Clf}_{\text{group}} \rrbracket = \lambda n_n \lambda P_{\langle e, t \rangle} \lambda x_e [*P(x) \wedge |||(P)(x) = n]$
- ▶ (23): representation of unit/group counting numeral structures

(23)



Definitions of $\#$ and $|||$

$$\begin{array}{ll}
 \forall n_n \forall P_{\langle e, t \rangle} \forall x_e [\#(P)(x) = n] & \leftrightarrow \exists Y_{\langle e, t \rangle} [Y \in \text{PAR}_c(x) \ \& \ \forall y_e [y \in Y \rightarrow y \in P \ \& \ |Y| = n]] \\
 \forall n_n \forall P_{\langle e, t \rangle} \forall x_e [|||(P)(x) = n] & \leftrightarrow \exists Y_{\langle e, t \rangle} [Y \in \text{PAR}_c(x) \ \& \ \forall y_e [y \in Y \rightarrow y \in *P \ \& \ |Y| = n]]
 \end{array}$$

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Pattern 2: abstract *Num+A*

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Unit vs. group counting

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Appendix

Pattern 1: Mandarin and Uzbek

Mandarin 'three'

san

san **tóu**_{unit}

san **qún**_{group}

Num

Num + **A**

Num + **B**

abstract

unit

group

Uzbek 'two'

ikki

ikki **dona**_{unit}

ikki **guruh**_{group}

Num

Num + **A**

Num + **B**

abstract

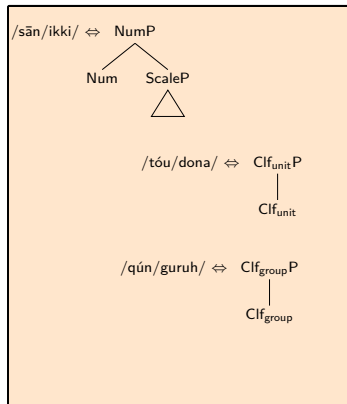
unit

group

Mandarin/Uzbek

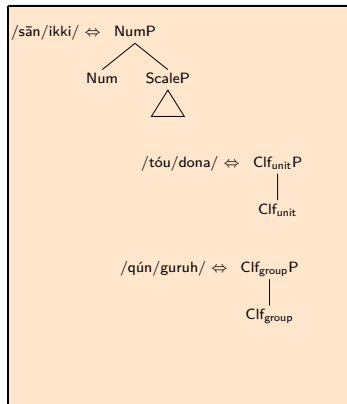
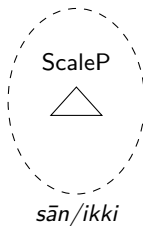
Abstract	Num
Object	Num-A
Group	Num-B

ScaleP



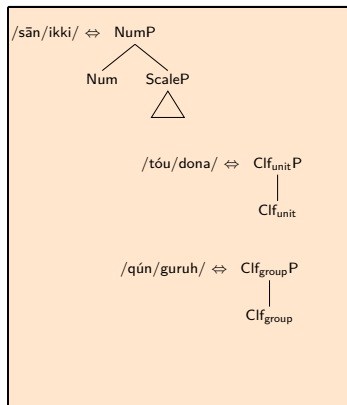
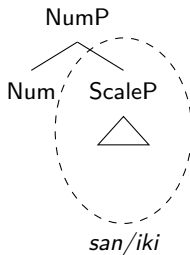
Mandarin/Uzbek

Abstract **Num**
Object Num-A
Group Num-B



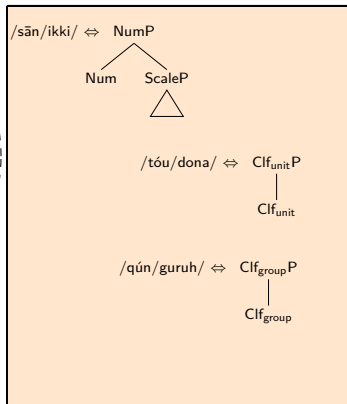
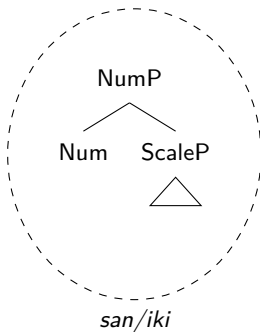
Mandarin/Uzbek

Abstract **Num**
Object Num-A
Group Num-B



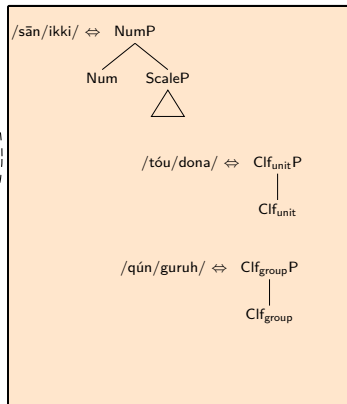
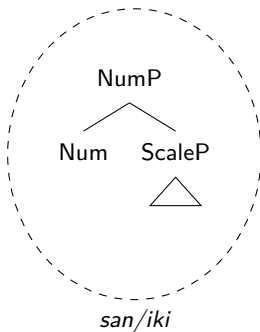
Mandarin/Uzbek

Abstract **Num**
Object Num-A
Group Num-B



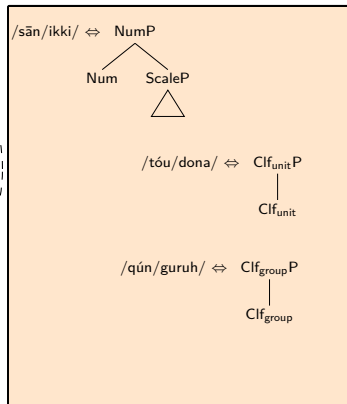
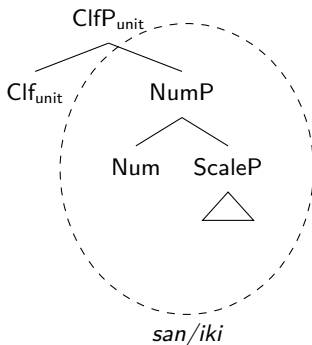
Mandarin/Uzbek

Abstract **Num**
Object Num-A
Group Num-B



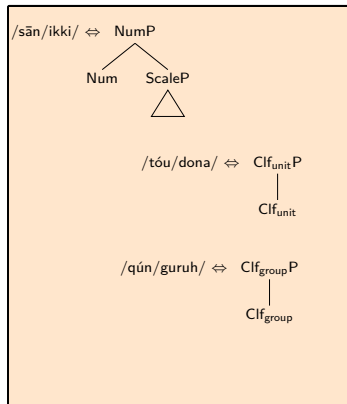
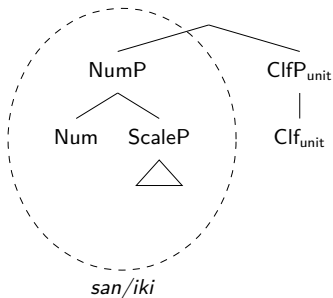
Mandarin/Uzbek

Abstract Num
Object **Num-A**
Group Num-B



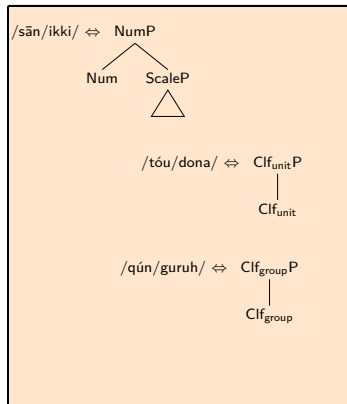
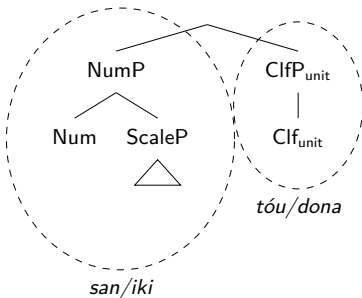
Mandarin/Uzbek

Abstract Num
Object **Num-A**
Group Num-B



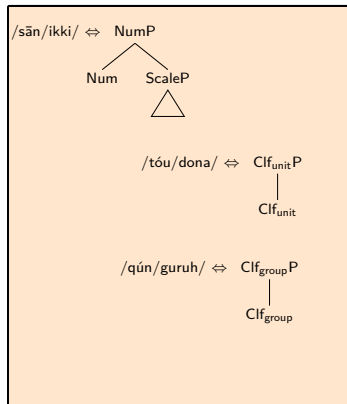
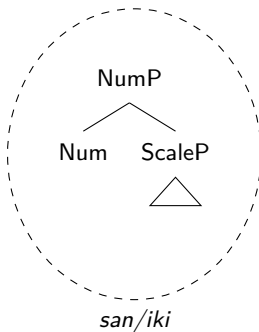
Mandarin/Uzbek

Abstract Num
Object **Num-A**
Group Num-B



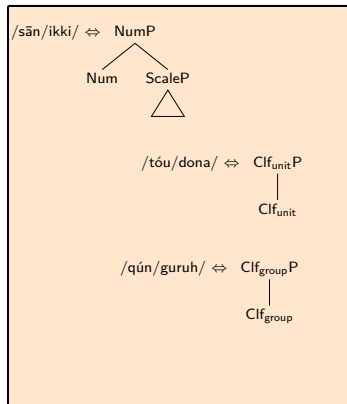
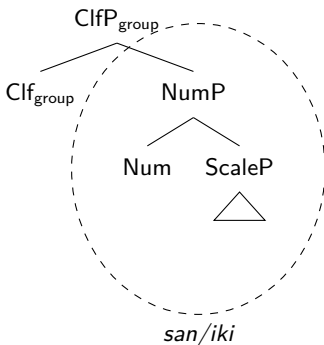
Mandarin/Uzbek

Abstract Num
Object Num-A
Group **Num-B**



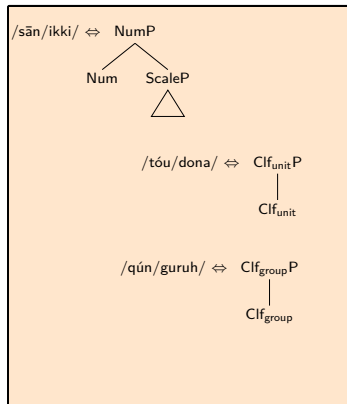
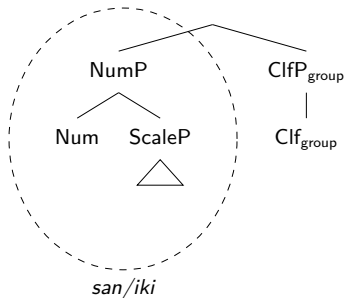
Mandarin/Uzbek

Abstract Num
Object Num-A
Group **Num-B**



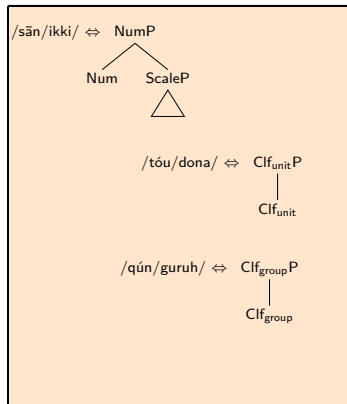
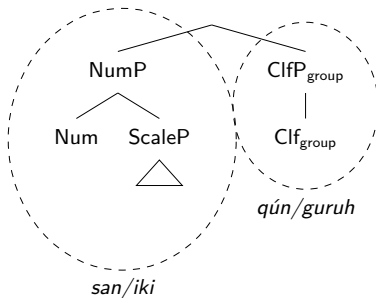
Mandarin/Uzbek

Abstract Num
Object Num-A
Group **Num-B**



Mandarin/Uzbek

Abstract Num
Object Num-A
Group **Num-B**



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Pattern 2: Yucatec Maya and Bwamu

Bwamu 'three'

bo tĩ

bo tĩ

gnũ-nɛ bo tĩ

Num + A

Num + A

Num + A + B

abstract

unit

group

Yucatec Maya 'one'

jum-p'éel

jum-p'éel

jum-p'éel cháach

Num + A

Num + A

Num + A + C

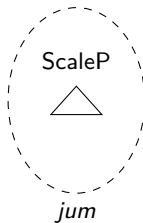
abstract

unit

group

Yucatec

Abstract **Num-A**
Object Num-A
Group Num-A-C



/jum/ ⇔ ScaleP



/p'éél/ ⇔ Clf_{unit}P

Clf_{unit}

NumP

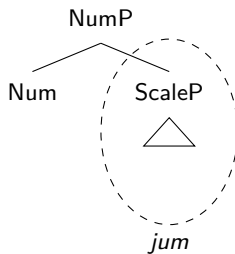
Num

/cháach/ ⇔ Clf_{group}P

Clf_{group}

Yucatec

Abstract **Num-A**
 Object Num-A
 Group Num-A-C



/jum/ $\Leftrightarrow$ ScaleP



/p'éel/ $\Leftrightarrow$ Clf_{unit}P

Clf_{unit}

NumP

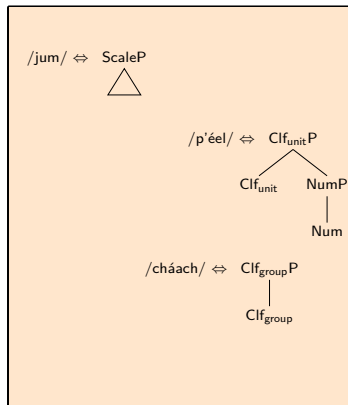
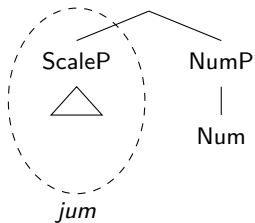
Num

/cháach/ $\Leftrightarrow$ Clf_{group}P

Clf_{group}

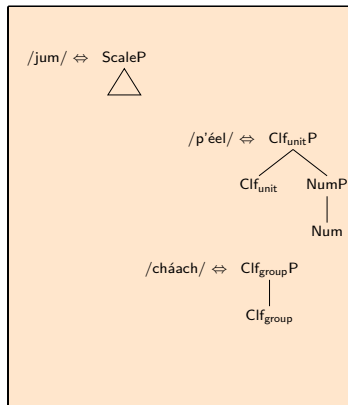
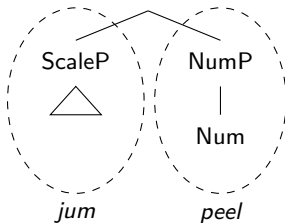
Yucatec

Abstract **Num-A**
 Object Num-A
 Group Num-A-C



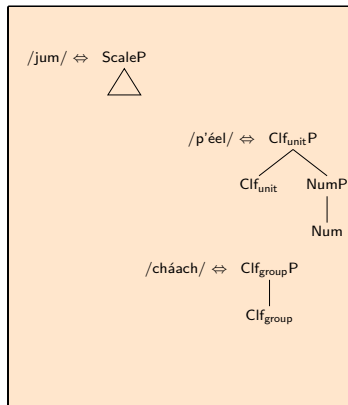
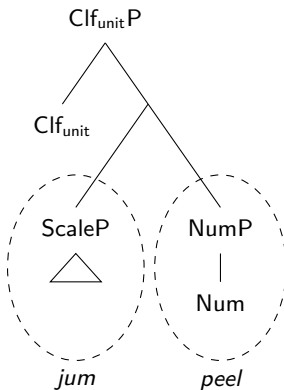
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Abstract **Num-A**
Object Num-A
Group Num-A-C



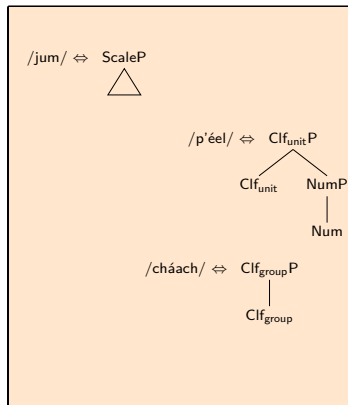
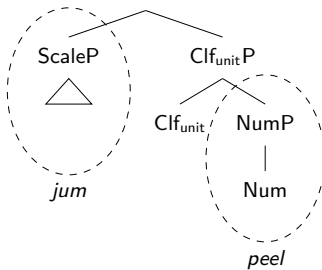
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Abstract Num-A
 Object **Num-A**
 Group Num-A-C



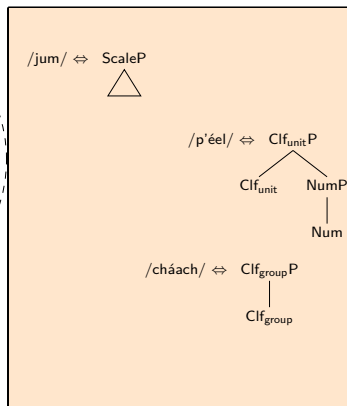
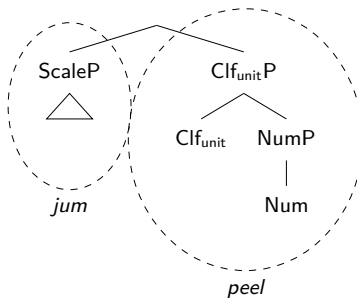
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Abstract Num-A
 Object **Num-A**
 Group Num-A-C



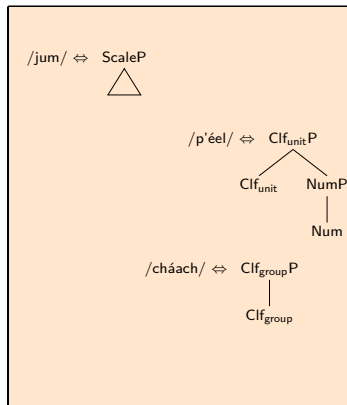
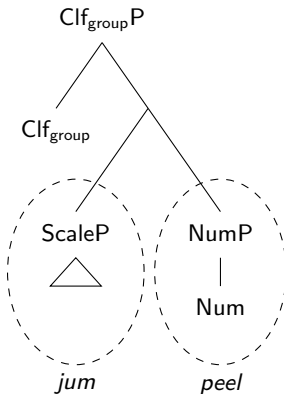
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Abstract Num-A
 Object **Num-A**
 Group Num-A-C



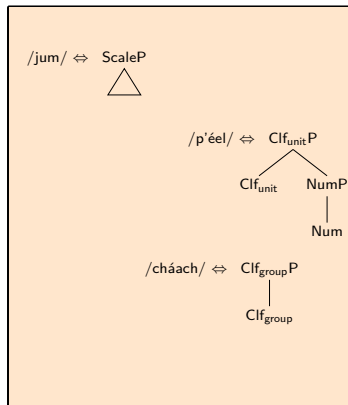
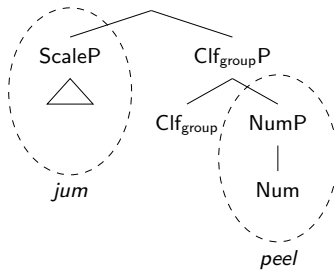
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Abstract Num-A
 Object Num-A
 Group **Num-A-C**



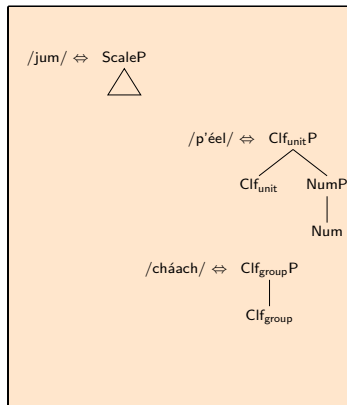
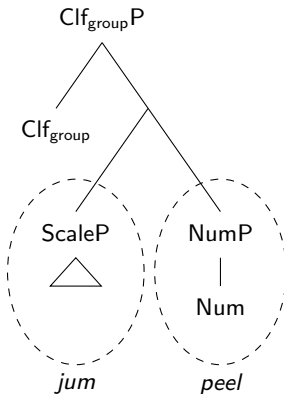
Yucatec

Abstract Num-A
 Object Num-A
 Group **Num-A-C**



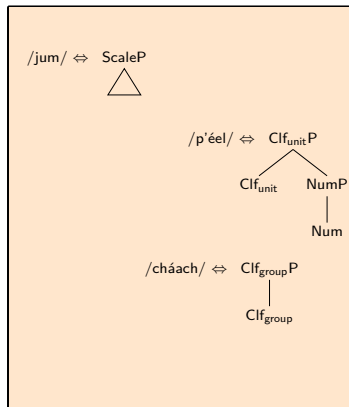
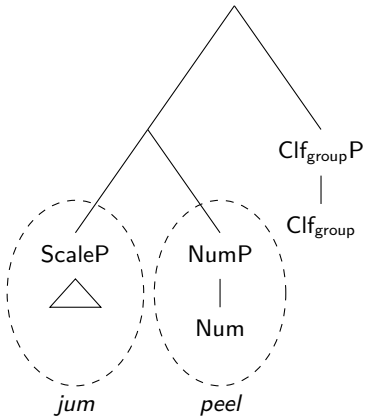
Yucatec

Abstract Num-A
 Object Num-A
 Group **Num-A-C**



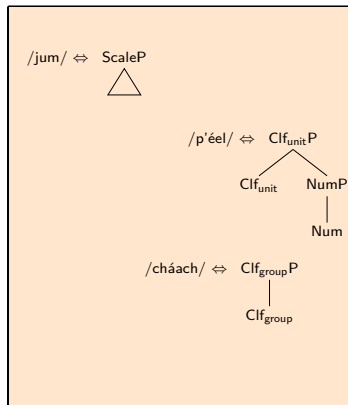
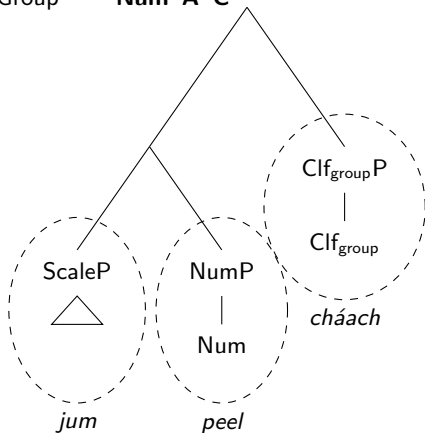
Yucatec

Abstract Num-A
 Object Num-A
 Group **Num-A-C**



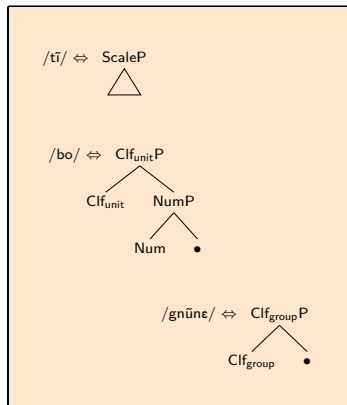
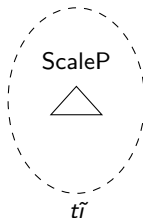
Yucatec

Abstract Num-A
 Object Num-A
 Group **Num-A-C**



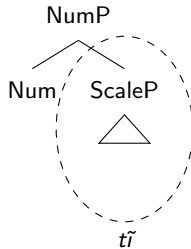
Bwamu

Abstract **Num-A**
 Object Num-A
 Group Num-A-C



Bwamu

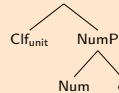
Abstract **Num-A**
Object Num-A
Group Num-A-C



/tĩ/ $\Leftrightarrow$ ScaleP



/bo/ $\Leftrightarrow$ Clf_{unit}P

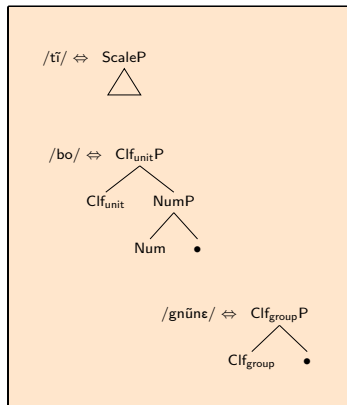
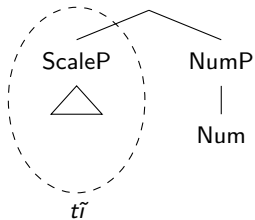


/gnũε/ $\Leftrightarrow$ Clf_{group}P



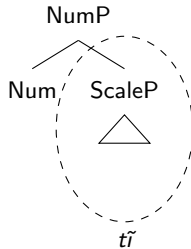
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Abstract **Num-A**
 Object Num-A
 Group Num-A-C



Bwamu

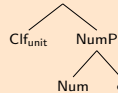
Abstract **Num-A**
 Object Num-A
 Group Num-A-C



$/t\tilde{i}/ \Leftrightarrow$ ScaleP



$/bo/ \Leftrightarrow$ Clf_{unit}P

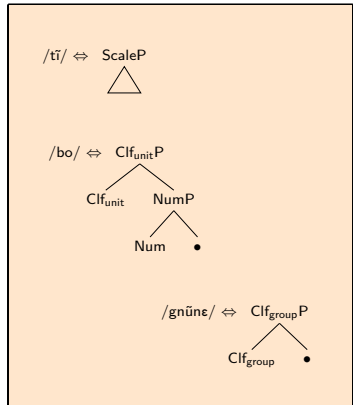
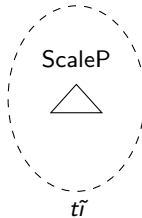
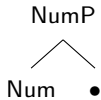


$/gnũ\epsilon/ \Leftrightarrow$ Clf_{group}P



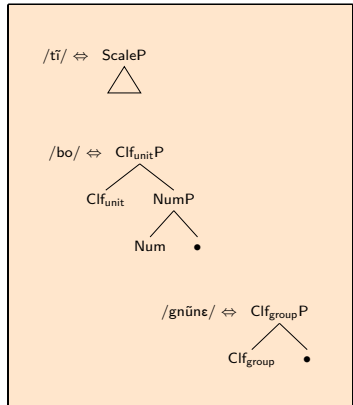
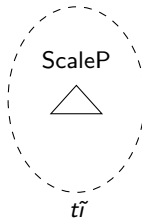
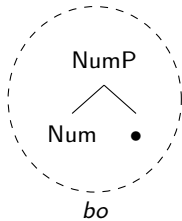
Bwamu

Abstract **Num-A**
 Object Num-A
 Group Num-A-C



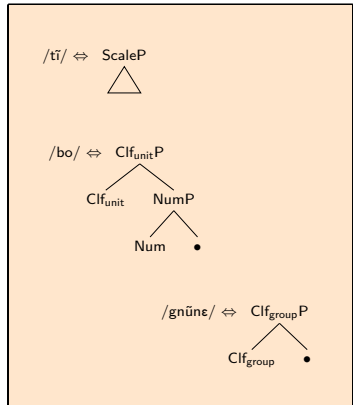
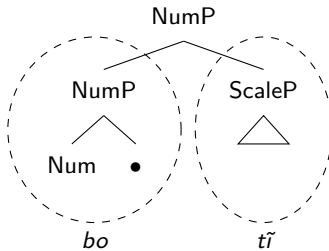
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Abstract **Num-A**
 Object Num-A
 Group Num-A-C



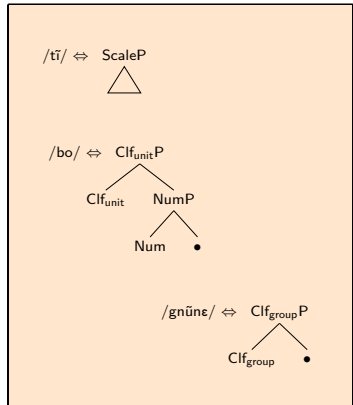
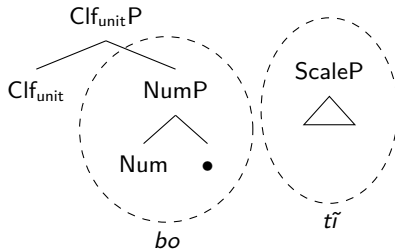
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Abstract **Num-A**
 Object Num-A
 Group Num-A-C



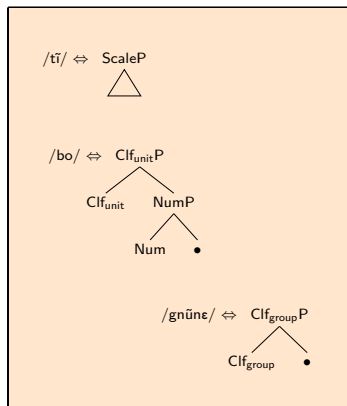
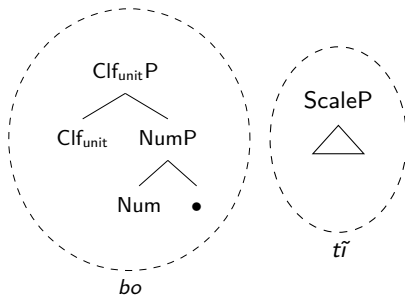
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Abstract Num-A
 Object **Num-A**
 Group Num-A-C



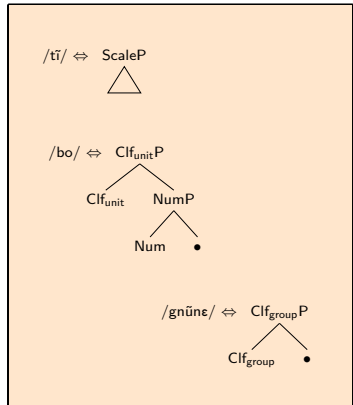
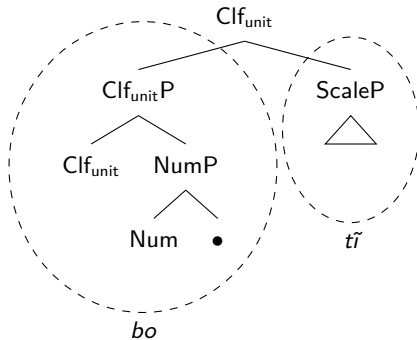
Bwamu

Abstract Num-A
 Object **Num-A**
 Group Num-A-C



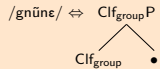
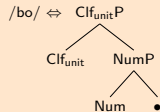
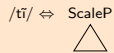
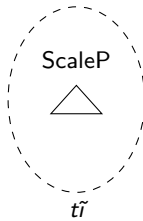
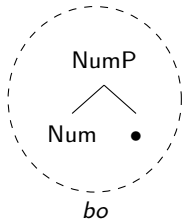
Bwamu

Abstract Num-A
 Object **Num-A**
 Group Num-A-C



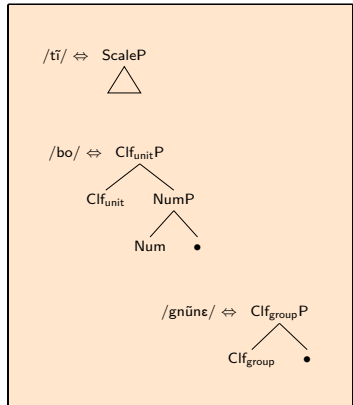
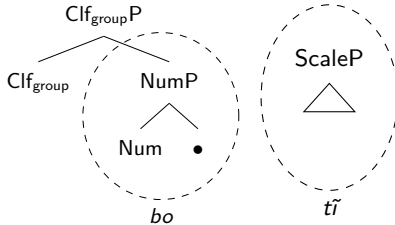
Bwamu

Abstract	Num-A
Object	Num-A
Group	Num-A-C



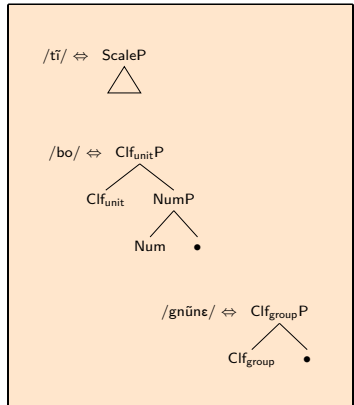
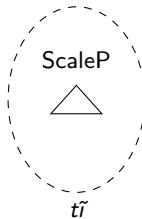
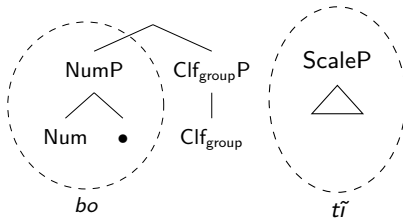
Bwamu

Abstract	Num-A
Object	Num-A
Group	Num-A-C



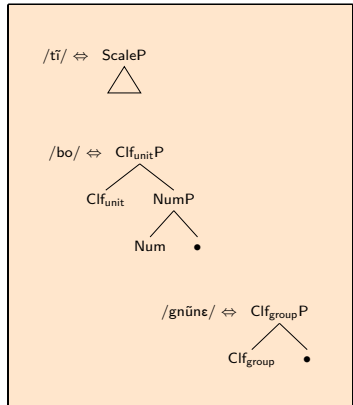
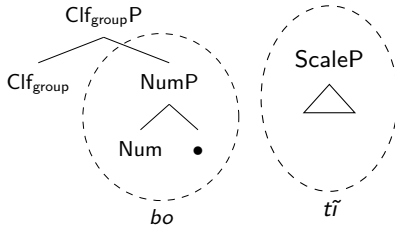
Bwamu

Abstract **Num-A**
 Object Num-A
 Group Num-A-C



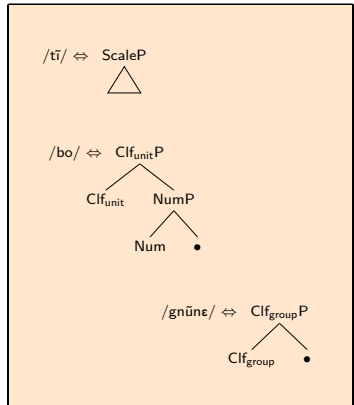
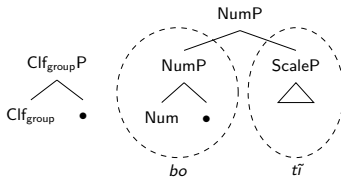
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Abstract **Num-A**
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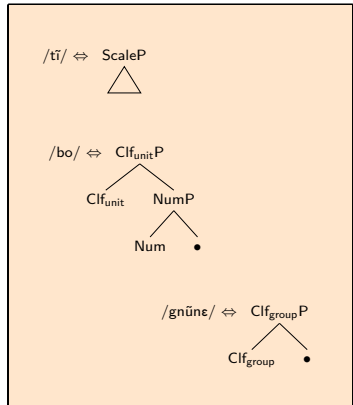
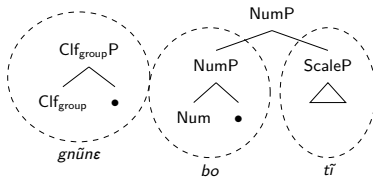
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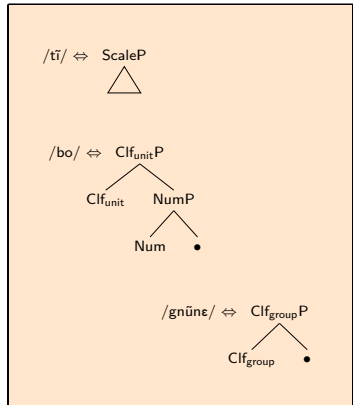
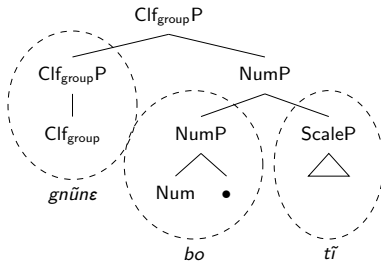
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Abstract **Num-A**
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Bwamu

Abstract **Num-A**
 Object Num-A
 Group Num-A-C



Outline

Unit vs. group counting

Pattern 1: unit $\rightarrow$ bimorphemic

Pattern 2: unit $\rightarrow$ bimorphemic

group $\rightarrow$ bimorphemic

group $\rightarrow$ tri-morphemic

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Abstract number concepts

Pattern 1: abstract bare *Num*

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Conclusion

Taking into account **group counting numeral constructions** provided insights into the following questions:

What is the relation between abstract number concepts and counting expressions?

- ▶ Numeral counting constructions properly contain an **abstract number core**
- ▶ Counting numerals: [CLF [**abstract-number**]]

How does natural language represent abstract number concepts?

- ▶ **Abstract number** concept expressions are **internally complex**.
- ▶ Number concept denoting constituent: [Num Scale]

Thank you!

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Appendix: Decomposition of Czech *troje*

Czech *tr-oj-e*

- (24) tř-i
'three'
- (25) tr-oj-e
'three groups of'
- (26) tr-oj-í
'three kinds of'
- (27) tr-oj-it-y
'triple (hamburger)'
- (28) tr-oj-násobn-ý
'three time (champion)'

Finnish *kah-de-t*

- (29) a. kak-si
'three'
b. kah-de-t
'three groups'
- (30) a. neljä 'four'
b. neljä-t
'four groups'
- (31) a. kolme 'three'
b. kolme-t
'three groups'
- (32) a. kissa
'cat'
b. kissa-t
cat-PL
'cats'

Alternative composition: *kahd-e-t* (thanks to Markus Pöchtrager)

Pattern 2: Czech and Finnish

Czech 'three'

tř-i

tř-i

tr-oj-e

Num + A

Num + A

Num + B + C

abstract

unit

group

Finnish 'two'

kak-si

kak-si

kah-de-t

Num + A

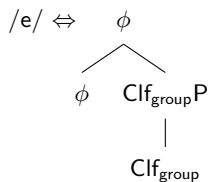
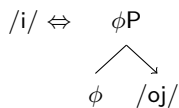
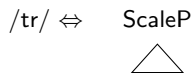
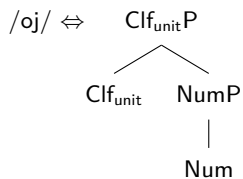
Num + A

Num + B + C

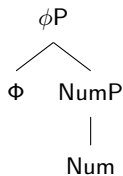
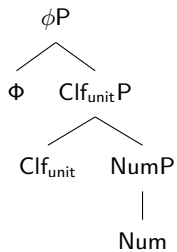
abstract

unit

group

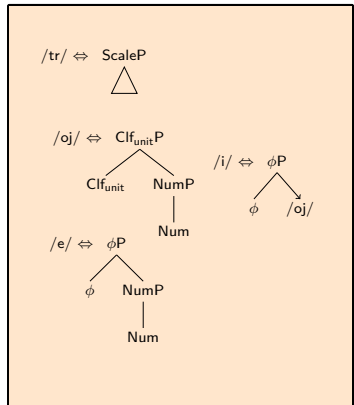
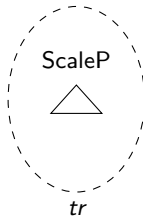


Possible lexicalisations for -i:



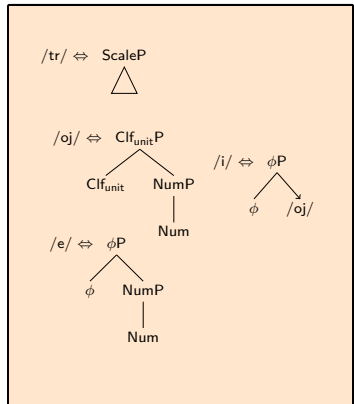
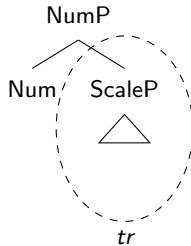
Czech / Finnish

Abstract	Num
Object	Num-A
Group	Num-B



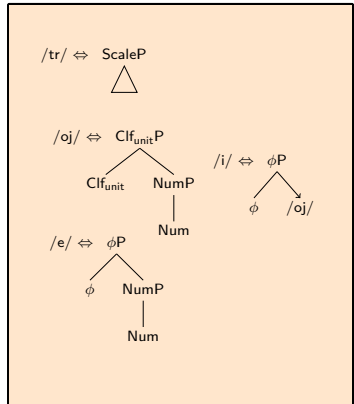
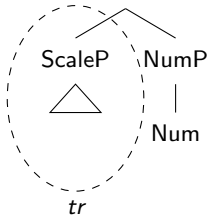
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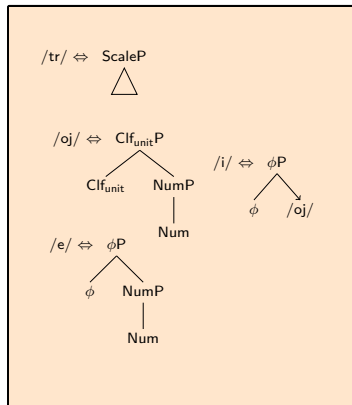
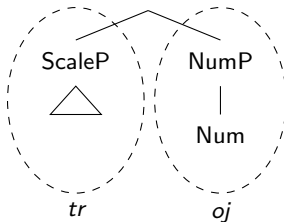
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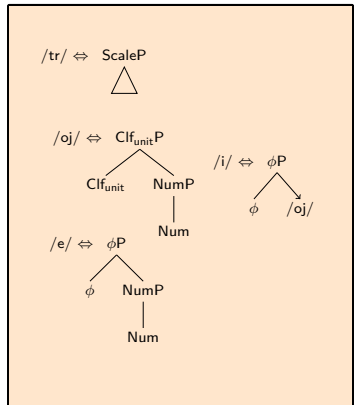
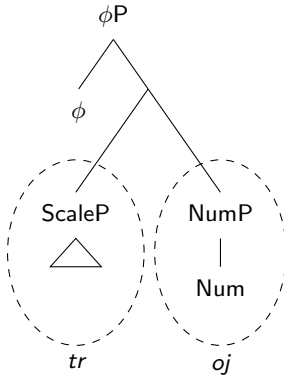
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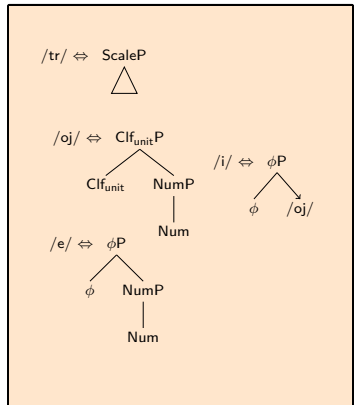
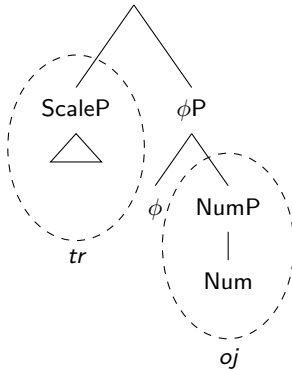
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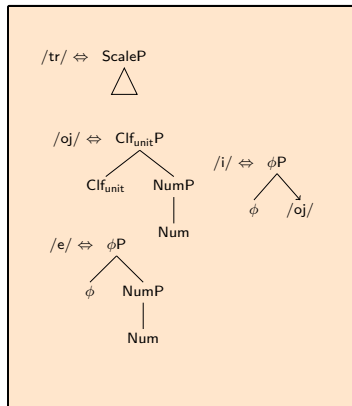
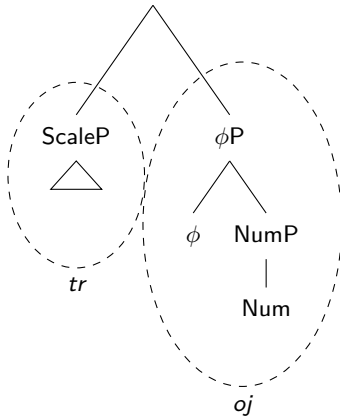
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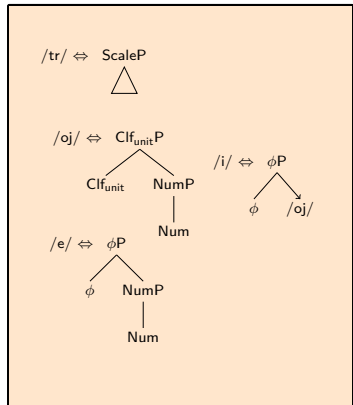
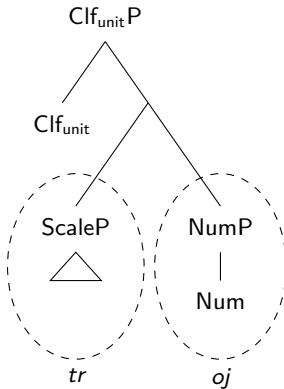
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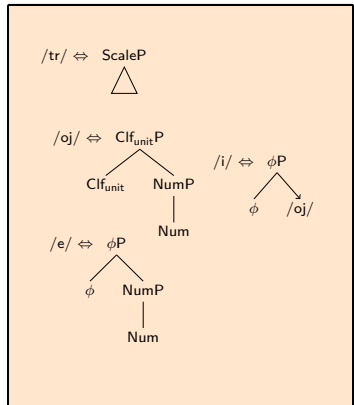
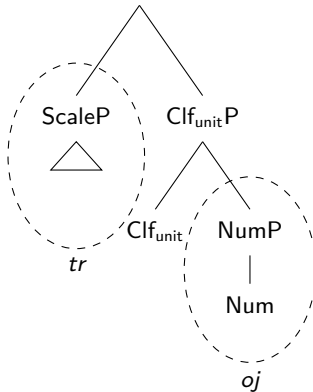
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Group Num-B



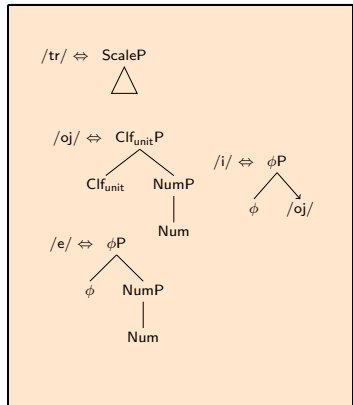
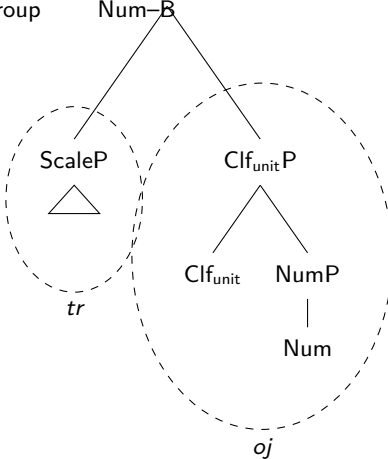
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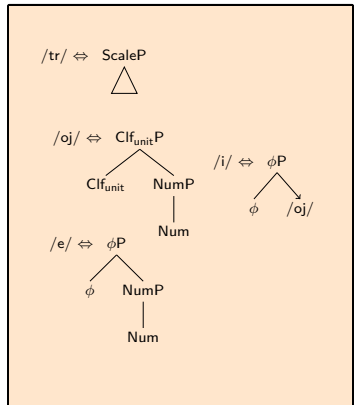
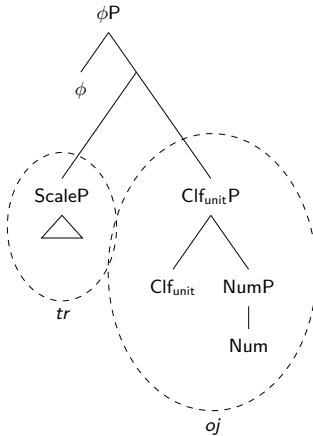
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Group Num-B



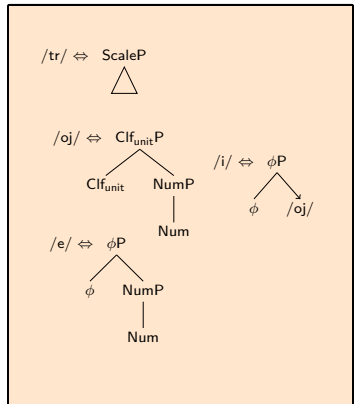
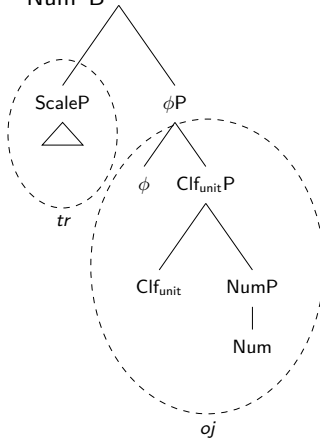
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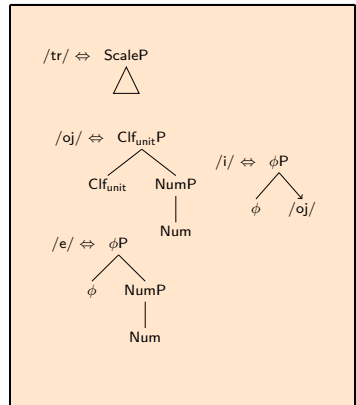
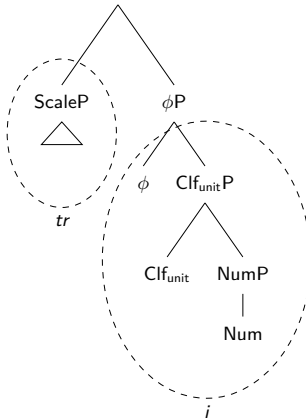
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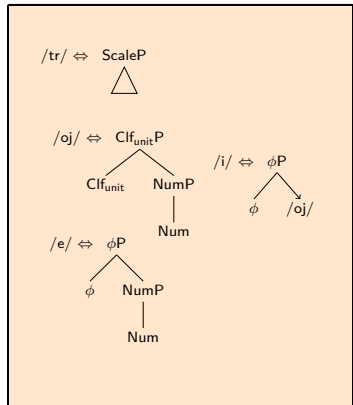
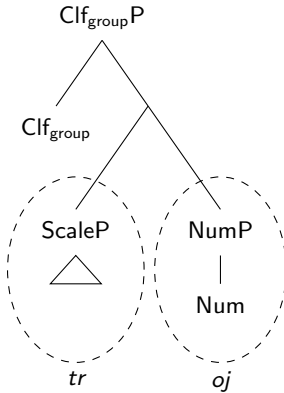
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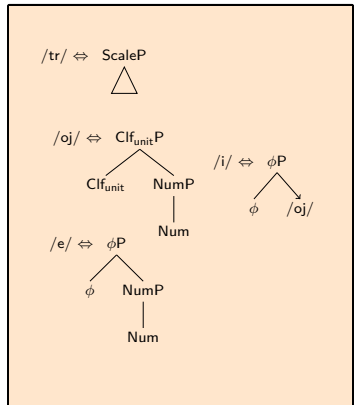
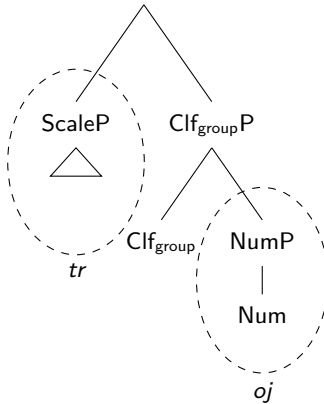
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Abstract Num
Object Num-A
Group **Num-C**



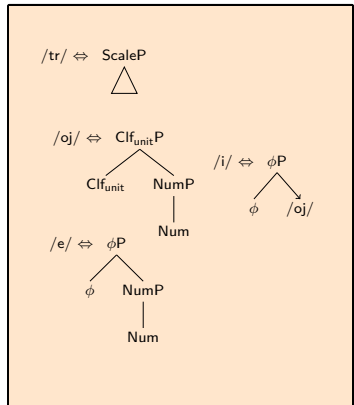
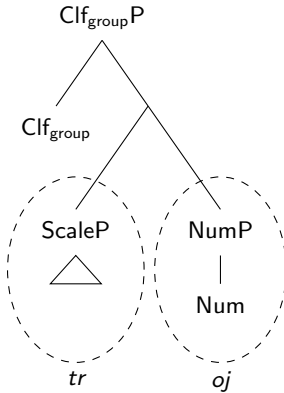
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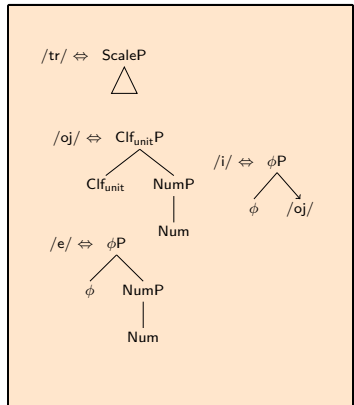
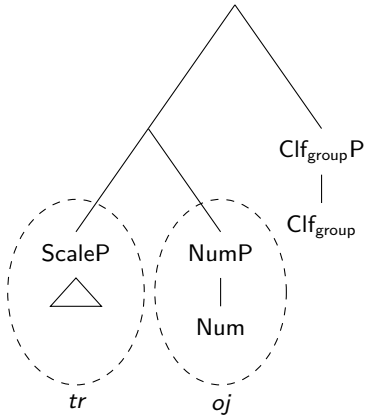
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Abstract Num
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Czech / Finnish

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